

React Without Surprises: Building Predictable Applications

Agenda

- State and Derived state
- Prop Drilling

Workshop at CityJS Athens 2026

- State and Derived state
- Prop Drilling
- Complex components
- Hooks anti-patterns
- Duplication (when to avoid it, and when not)
- You might not need useEffect
- And many more...

Housekeeping

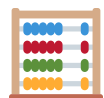
Feel free to interrupt me for:

- questions
- relevant comments



All code examples are:

- Real-world (realistic) examples
- Fully functional
- Production ready



What is derived state?

- Values **computed** / **calculated** from existing state
- No need for separate `useState` — just derive it!
- No need for `useEffect` syncs

Why use derived state

- Easier to understand code (single source of truth)
- Less state (`useState`) and side effects (`useEffect`)
- Cleaner components and UI
- Prevents bugs from out-of-sync state
- Better performance (fewer re-renders)

Live demo: Todo App

Live demo: Profile Form

Rule of thumb

If you can calculate it from existing state/props,
don't store it in state

🕒 What is prop drilling?

- Passing props through **multiple component layers**
- Intermediate components don't use the props — just forward them
- Creates tight coupling and maintenance nightmares

```
App → Header → HeaderActions → UserInfo → Profile
      ↓ user    ↓ user          ↓ user    ↓ user
```

Common signs of prop drilling

- Props passed through 3+ component levels
- Components accepting props they don't use
- Adding a new prop requires editing many files
- Renaming a prop becomes a refactoring nightmare
- "Why does this component need `user`?"

How to fix it

- **Component composition** (children pattern)
- **React Context API** (built-in)
- **Global state management libraries** (Zustand, Jotai, Redux)

Live demo: App layout

When to use Global state vs Props

Use Props	Global state
1-2 levels deep	3+ levels deep
Few consumers	Many consumers
Explicit data flow	Shared/global data

Key takeaways

1. **Derive, don't store** — if it can be calculated, don't `useState` it
2. **Prop drilling is a code smell** — fix it early before it spreads

 **That's all folks**

Questions / Discussions?